

FUNDAMENTALS

- $A' = A^c = A$ complement
- $P[A^c] = 1 - P[A]$
- $P[A \cup B] = P[A] + P[B] - P[A \cap B]$
- Mutually exclusive if $P[A \cap B] = 0$

CONDITIONAL PROBABILITY

- $P[A | B] = \frac{P[A \cap B]}{P[B]}$
- $P[A \cap B] = P[A] \cdot P[B | A]$
 $= P[B] \cdot P[A | B]$
- A and B are independent means:
 - $P[A \cap B] = P[A] \cdot P[B]$ (definition)
 - $P[A | B] = P[A]$ (implied)
- A , B , and C are independent if every possible combination factors
 For example:
 - $P[ABC'] = P[A]P[B]P[C']$
 - $P[A^cB] = P[A^c]P[B]$
- $A_1, A_2, \dots, A_k$ are a partition if:
 - $A_i \cap A_j = \emptyset, i \neq j$
 - $\sum P[A_i] = 1$
- If $A_1, A_2, \dots, A_k$ are a partition:
 - Law of total probability:
 $P[B] = \sum P[B | A_i] \cdot P[A_i]$
 - Bayes' Theorem:
 $P[A_i | B] = \frac{P[A_i \cap B]}{P[B]}$
 $= \frac{P[A_i \cap B]}{\sum P[B | A_j] \cdot P[A_j]}$

VARIANCES

- $\text{Var}[X] = \sigma^2 = E[X^2] - (E[X])^2$
- $E[X^2] = \text{Var}[X] + (E[X])^2$
- $\text{Var}[cX] = c^2 \text{Var}[X]$
- $\text{SD}[X] = \sqrt{\text{Var}[X]}$
- Coefficient of Variation = $\frac{\text{SD}[X]}{E[X]}$

CDFs AND DENSITIES

- $F(x) = P[X \leq x] = \text{CDF}$
 - Discrete X :

$$F(x) = \sum_{n \leq x} P[X = n]$$
 - Continuous X :

$$F(x) = \int_{-\infty}^x f(t) dt$$
- Density $f(x) = \frac{d}{dx} F(x)$
- Mixed distributions:
 - If $F(x)$ has a jump at a ,
 e.g., $F(a) > \lim_{x \uparrow a} F(x) = F(a^-)$
 Then $P[X = a] = F(a) - F(a^-)$

MOMENTS

- Discrete:
 - $E[X] = \mu = \sum_x x \cdot P[X = x]$
 - $E[X^2] = \sum_x x^2 \cdot P[X = x]$
 - $E[g(X)] = \sum_x g(x) \cdot P[X = x]$
- Continuous:
 - $E[X] = \mu = \int x \cdot f(x) dx$
 - $E[X^2] = \int x^2 \cdot f(x) dx$
 - $E[g(X)] = \int g(x) \cdot f(x) dx$
- Mixed:
 - Find discrete / continuous pieces, then sum
- Survival approach: If $X \geq 0$ then:
 - $E[X] = \int_0^\infty S(x) dx$
 - $E[g(X)] = \int_0^\infty g'(x) S(x) dx$
 assuming $g(0) = 0$

JOINT DISTRIBUTIONS

- $P[X = x] = \sum_y P[X = x, Y = y]$
 = Marginal distribution of X
- $P[X = x | Y = y]$

$$= \frac{P[X = x, Y = y]}{P[Y = y]}$$

 = Conditional distribution of X
- $F_{X,Y}(x, y) = P[X \leq x, Y \leq y]$
- $F_X(x) = P[X \leq x] = F_{X,Y}(x, \infty)$

COVARIANCES

- $\text{Cov}[X, Y] = E[XY] - E[X] \cdot E[Y]$
- $\text{Cov}[X, X] = \text{Var}[X]$
- $\text{Cov}[aX + bY, cZ + dW]$

$$= ac \text{Cov}[X, Z] + ad \text{Cov}[X, W] + bc \text{Cov}[Y, Z] + bd \text{Cov}[Y, W]$$
- $\text{Var}[aX + bY] =$

$$a^2 \text{Var}[X] + 2ab \text{Cov}[X, Y] + b^2 \text{Var}[Y]$$
- $\text{Var}[\sum X_i]$

$$= \sum \text{Var}[X_i] \quad \text{if independent}$$
- $\text{Corr}[X, Y] = \frac{\text{Cov}[X, Y]}{\text{SD}[X] \text{SD}[Y]}$

CONDITIONAL MOMENTS

- $E[X | Y = y]$

$$= \sum_x x \cdot P[X = x | Y = y]$$
- $E[X] = E[E[X | Y]]$

ORDER STATISTICS

- $X_1, X_2, \dots, X_n$ iid random variables,
 $Y_1 \leq Y_2 \leq \dots \leq Y_n$ sorted version
- $P[\min\{X_1, \dots, X_n\} > x]$

$$= P[Y_1 > x] = (P[X > x])^n$$
- $P[\max\{X_1, \dots, X_n\} \leq x]$

$$= P[Y_n \leq x] = (P[X \leq x])^n$$
- $f_{Y_i}(y) = \text{density of } Y_i =$

$$i \cdot \binom{n}{i} [F_X(y)]^{i-1} f_X(y) [1 - F_X(y)]^{n-i}$$

DISCRETE DISTRIBUTIONS

Variable	Key Properties	$P[X = x]$	$E[X]$	$\text{Var}[X]$
Uniform on $\{1, 2, \dots, n\}$		$\frac{1}{n}, 1 \leq x \leq n$	$\frac{n+1}{2}$	$\frac{n^2-1}{12}$
Bernoulli	0 or 1 $1-p, x=0$	$p, x=1$	p	$p(1-p)$
Binomial	Sum of Bernoullis number of successes in n trials	$\binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$
Geometric on $\{1, 2, \dots\}$	Number of trials until first success	$p(1-p)^{x-1}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Geometric on $\{0, 1, \dots\}$	Number of failures before first success	$p(1-p)^x$	$\frac{1}{p} - 1$	$\frac{1-p}{p^2}$
Negative Binomial	Sum of Geometric on $0, \dots$ Number of failures until r -th success	$\binom{x+r-1}{r-1} p^r (1-p)^x$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$
Poisson	Sum of Poissons is Poisson	$e^{-\lambda} \frac{\lambda^n}{n!}$	λ	λ

Bernoulli Variance Shortcut: If X is a random variable that can only take on 2 values, with $P[X = a] = p$ and $P[X = b] = 1 - p$, then $\text{Var}(X) = (b - a)^2 p(1 - p)$.

CONTINUOUS DISTRIBUTIONS

Densities are only given for the region on which they are positive, and are 0 otherwise.

Distribution	range	density	cdf	$E[X]$	$\text{Var}[X]$
Uniform on $(0, 1)$	$0 < x < 1$	1	x for $0 < x < 1$	$\frac{1}{2}$	$\frac{1}{12}$
Uniform on (a, b)	$a < x < b$	$\frac{1}{b-a}$	$\frac{x-a}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential with "rate" λ and mean $\theta = \frac{1}{\lambda}$	$0 < x < \infty$	$\lambda e^{-\lambda x} = \frac{1}{\theta} e^{-x/\theta}$	$1 - e^{-\lambda x} = 1 - e^{-x/\theta}$	$\theta = \frac{1}{\lambda}$	$\theta^2 = \frac{1}{\lambda^2}$
Gamma	$0 < x < \infty$	$\frac{x^{\alpha-1} e^{-x/\theta}}{\theta^\alpha (\alpha-1)!}$	see below	$\alpha\theta$	$\alpha\theta^2$
Standard Normal	$-\infty < x < \infty$	$\frac{1}{\sqrt{2\pi}} e^{-x^2/2}$	$\Phi(x)$	0	1
Normal(μ, σ^2)	$-\infty < x < \infty$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$	$\Phi\left(\frac{x-\mu}{\sigma}\right)$	μ	σ^2

GAMMA SHORTCUT

One trick to remembering the Gamma distribution is to recall that if α is an integer, then a $\text{Gamma}(\alpha, \theta)$ distribution is the sum of α exponentials with mean θ .

- For $\alpha = 1$, a Gamma is an exponential with mean θ and so has CDF $P[X \leq t] = 1 - e^{-t/\theta}$
- For $\alpha = 2$, a Gamma has CDF

$$P[X \leq t] = 1 - e^{-t/\theta} - \left(\frac{t}{\theta}\right) e^{-t/\theta}$$

- For $\alpha = 3$, a Gamma has CDF

$$P[X \leq t] = 1 - e^{-t/\theta} - \left(\frac{t}{\theta}\right) e^{-t/\theta} - \left(\frac{(t/\theta)^2}{2}\right) e^{-t/\theta}$$

- Note that each time we increase α by 1, we subtract another term from the CDF that resembles a term from the probability distribution of a Poisson. In general, if α is an integer, the CDF of a $\text{Gamma}(\alpha, \theta)$ is given by:

$$P[X \leq t] = P[X < t] = 1 - P[N \leq \alpha - 1] = P[N \geq \alpha]$$

where N is a Poisson random variable with mean $\lambda = t/\theta$.